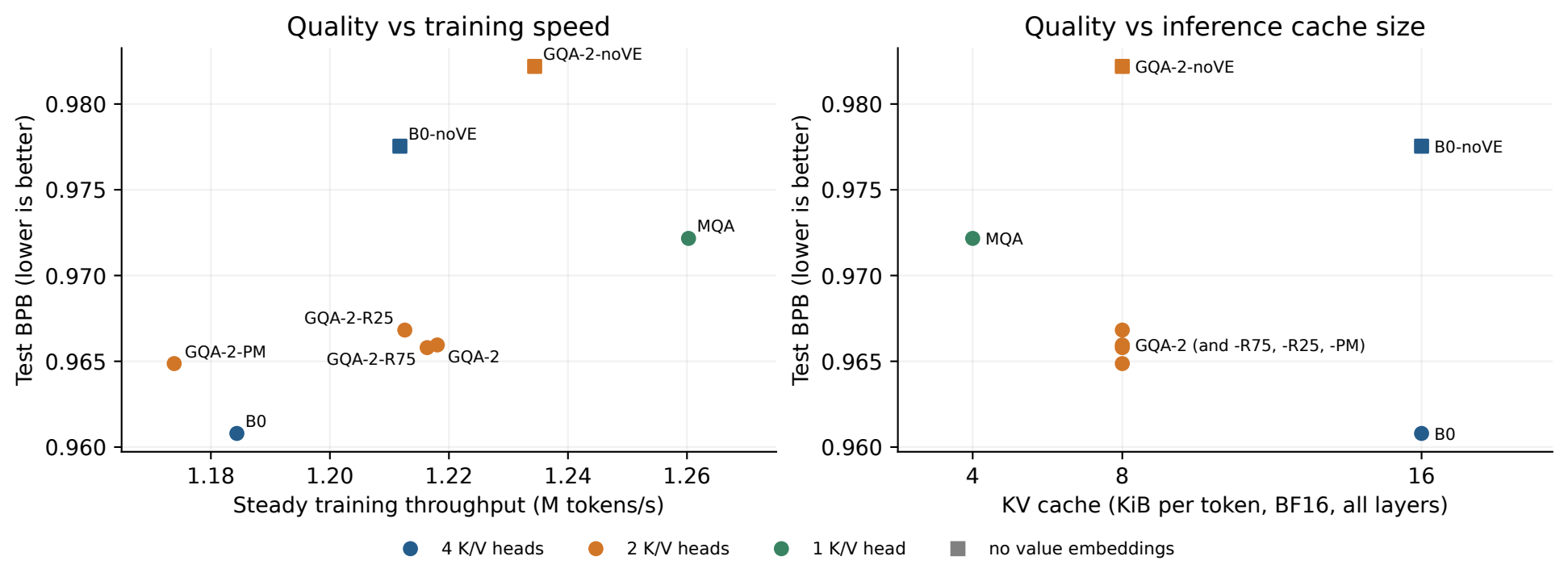
Quality vs training speed
Test BPB (lower is better)
0.980
0.975
0.970
0.965
0.960
1.18
1.20
1.22
1.24
1.26
Steady training throughput (M tokens/s)
GQA-2-noVE
B0-noVE
MQA
GQA-2-R25
GQA-2-PM
GQA-2-R75
GQA-2
B0
Quality vs inference cache size
Test BPB (lower is better)
0.980
0.975
0.970
0.965
0.960
4
8
16
KV cache (KiB per token, BF16, all layers)
GQA-2-noVE
B0-noVE
MQA
GQA-2 (and -R75, -R25, -PM)
B0
4 K/V heads
2 K/V heads
1 K/V head
no value embeddings